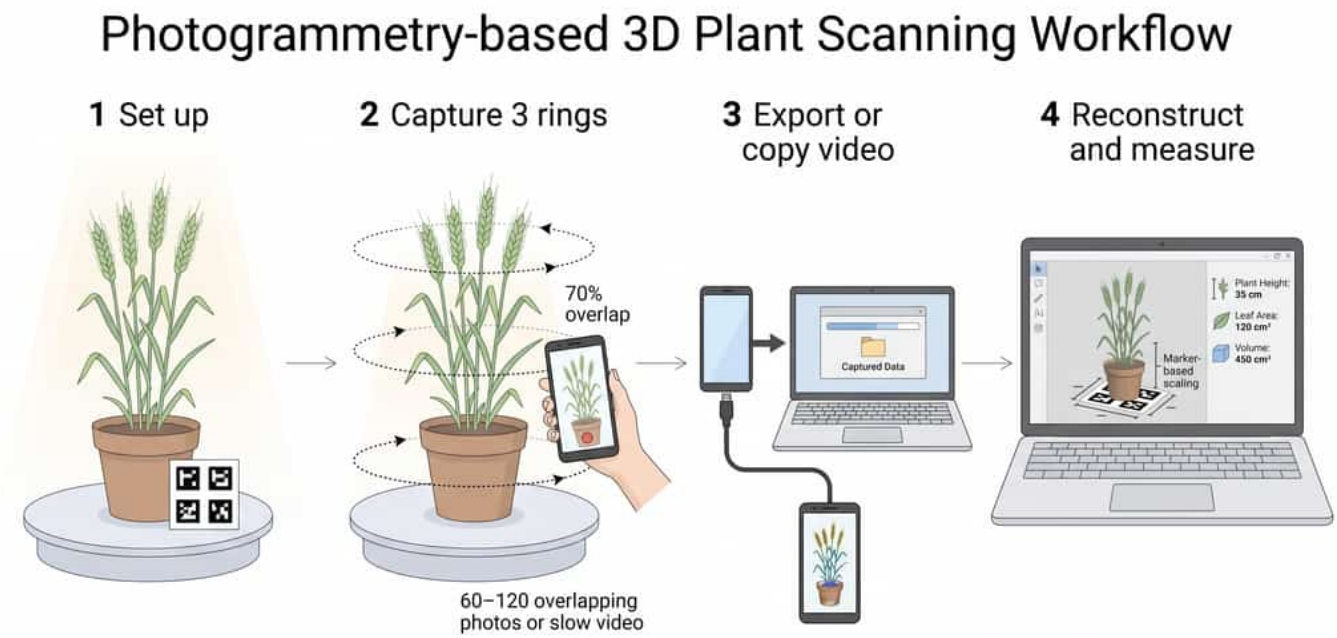
# SOP-2 Android phone

Photogrammetry from photos or video, or an app export, scaled with ArUco markers

SOP-2 · PhenoForge capture SOP · version 1.0 (October 2026)




## What you need

**Checklist**

- Any recent Android phone with a good main camera (4K video)
- Option A: a scanning app that exports OBJ / GLB / PLY or Gaussian-splat PLY (e.g. Polycam, Scaniverse, KIRI Engine, RealityScan)
- Option B: just the camera app (photos or video) – PhenoForge reconstructs with COLMAP
- ArUco card (50 mm markers), turntable optional
- USB cable to copy files

## Capture

**1 Lock the camera**
Turn off HDR, beauty and AI scene modes. Use the main (1×) lens only. Tap and hold on the plant to lock focus and exposure (AE/AF lock).

**2 Capture three rings**
Photos: 20–40 per ring with ~70 % overlap, at low, middle and top-down heights (60–120 in total). Video: 4K 30 fps, one slow circle per ring, ~40 s per circle, no fast turns.

**3 Keep the markers in view**
The ArUco card must appear in most photos, flat and sharp.

**4 Export or copy**
Option A: export the app's model as OBJ/GLB/PLY. Option B: copy the photo folder or the video to the PC by USB (DCIM/Camera).

## In PhenoForge

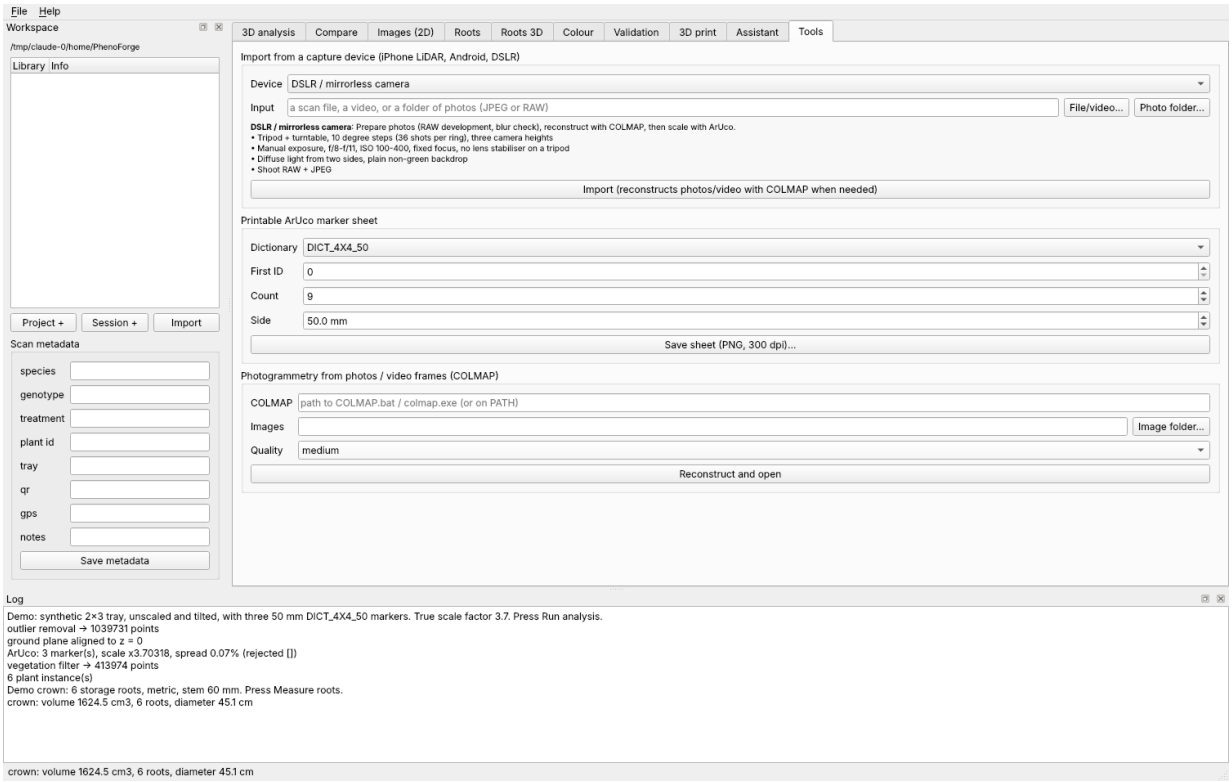
**1** **Import**
**Tools > Import from a capture device**, device *Android phone*. Choose the app export, the video (**File/video…**) or the photo folder (**Photo folder…**). For video the sharpest ~120 frames are picked automatically; blurred or over-exposed photos are rejected and listed in the log.

**2** **Reconstruct**
Photos and video are reconstructed with COLMAP (set its path once in Tools; install from colmap.github.io). Use quality *medium* first; *high* needs an NVIDIA GPU and more time.

**3** **Scale with ArUco**
The model opens in the 3D analysis tab with scale source **ArUco markers** preselected. Type the measured marker size.

**4** **Segment**
Section 3: *single* for one plant, *watershed* for a tray of touching plants (enter the number of plants), or *grid* with rows, columns and pot pitch for a bench of equally spaced pots.

**5** **Run and export**
Press **Run analysis**. Results appear in the Scan / Per plant / Leaves tables; **Save results** writes JSON and CSV. Colour the view by Height, Greenness or Plants to check the segmentation.



Tools tab: device import with capture tips for the chosen device.

## Troubleshooting

| Symptom | Likely cause | Fix |
|---|---|---|
| COLMAP registers few images | Too little overlap or motion blur | Move slower; more photos per ring; better light |
| Plant is doubled or smeared | Plant moved (wind, touching) | Capture indoors; do not touch the pot |
| No markers detected | Card curved, too small or out of view | Flat card, 50 mm markers, in view from all sides |
| Scale differs between markers | Marker not flat or partly hidden | Rejected automatically above 4 %; reposition the card |